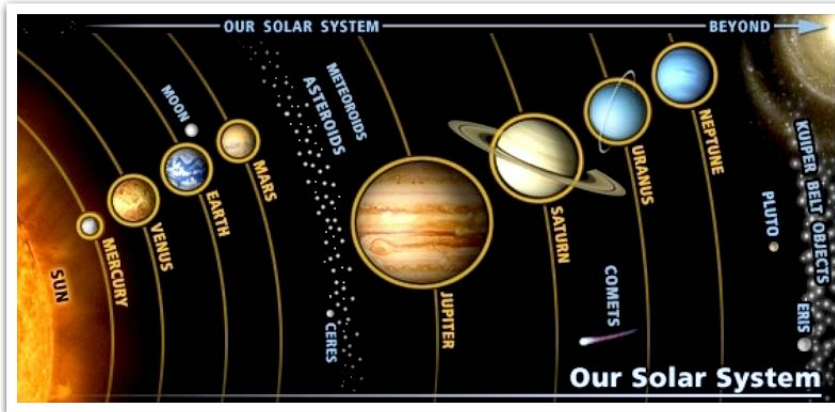


UPCAT Review: Science I (General Science and Biology)

A. Review of General Science

I. Universe, Astronomy and Solar System

1. Solar System



2. Kepler's Laws of Planetary Motion

- The orbit of a planet is an ellipse with the Sun at one of the two foci.
- A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.
- The square of the orbital period of a planet is proportional to the cube of the semi-major axis of its orbit.

II. Geology

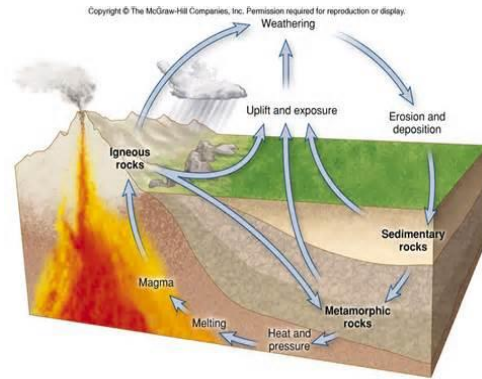
1. Rocks and Minerals

a. Classification of Rocks

- Igneous:** forms from cooled magma or lava.
 - Intrusive igneous rock (ex. granite) - forms in underground pockets
 - Extrusive igneous rock (ex. obsidian) - forms above ground
- Sedimentary:** Erosion from water and weather carries deposits downstream which will form layers. As pressure accumulates over thousands of years, the lower levels are pressed into sedimentary rock.
 - Clastic sedimentary rocks (ex. sandstone) - formed from inorganic debris.
 - Organic sedimentary rocks (ex. limestone or coal) - formed from plant or animal remains
 - Chemical sedimentary rock (ex. rock salt) - formed when dissolved materials come out of solution.
- Metamorphic:** Form when other types of rocks undergo extreme heat, pressure and chemical processes deep in Earth's crust.
 - Foliated metamorphic rocks (ex. slate) form with visible layers or banded striations from pressure.
 - Non-foliated metamorphic rocks (ex. marble) form more irregular striations

- b. **Rock Cycle:** Each form of rock can turn into the other under the right circumstances, thereby continuing the ever-shifting rock cycle.

- i. Igneous and sedimentary rock → metamorphic rock: high pressure and heat
- ii. Sedimentary and metamorphic rock → igneous rock: melting back into magma
- iii. Igneous and metamorphic rock → sedimentary rock: erosion



c. **Elements on the Earth's Crust**

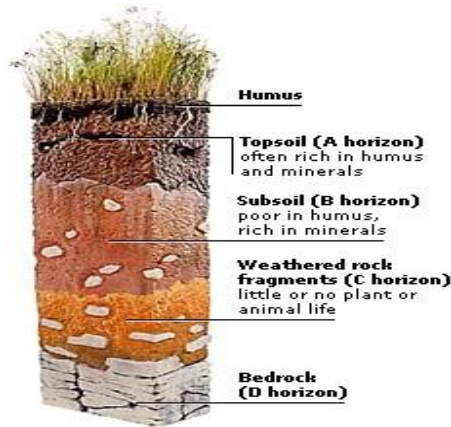
Element name	Symbol	% weight on Earth's crust
Oxygen	O	47
Silicon	Si	28
Aluminum	Al	8
Iron	Fe	5
Calcium	Ca	3.5
Sodium	Na	3
Potassium	K	2.5
Magnesium	Mg	2
All other elements		1

Source:

<http://www.rsc.org/Education/Teachers/Resources/jesei/minerals/students.htm>

- d. **Minerals:** substances formed naturally in the Earth with definite chemical composition and structure. There are over 3000 minerals known. Some are rare and precious such as diamond, while others are more ordinary, such as quartz.
- e. **Weathering:** one of the destructive processes on earth (together with Erosion and Transportation); exposes most rocks to a different physio-chemical environment from where they formed.
- i. **Mechanical weathering** - which produces clasts of rock/mineral which are smaller than the original, and of the same general composition. There are two basic processes:
 1. Fracturing, which is like a hammer and produces angular clasts
 2. Abrasion, which works like sandpaper and rounds off the rough edges.
 - ii. **Chemical weathering** - also produces smaller pieces, but of a different chemical composition - the minerals have been changed in an attempt to be more in balance with surface conditions. The most common product of chemical weathering is clay.
 - iii. **Biological weathering** - really isn't a type of its own; the actions of plants (and animals) can and do lead to both mechanical and biological weathering.

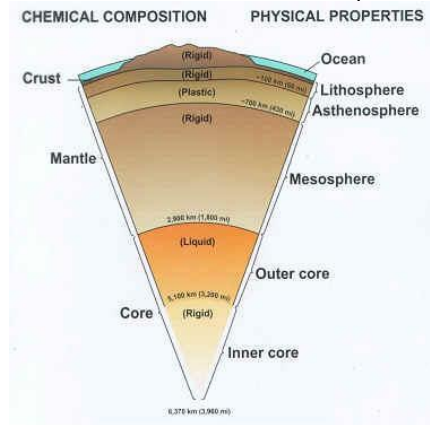
f. Layers of Soil



g. Mohs' Scale of Hardness

Mineral	Mohs relative Hardness	Scratch Test	Rosiwal absolute Hardness	Vickers kp/mm^2
Talc	1	scrapeable with fingernail	0.03	2.4
Gypsum	2	scratches with fingern.	1.25	36
Calcite	3	scr. with copper coin	4.5	109
Fluorite	4	easily scr. with knife	5	189
Apatite	5	still scr. with knife	6.5	536
Orthoclase	6	scr. with steel file	37	795
Quartz	7	scratches window glass	120	1,120
Topaz	8	scratches quartz	175	1,427
Corundum	9	scratches topaz	1,000	2,060
Diamond	10	scratches corundum	140,000	10,060

2. Earth's Surface (earthquakes, volcanoes, faults, ,mountains, plate tectonics)



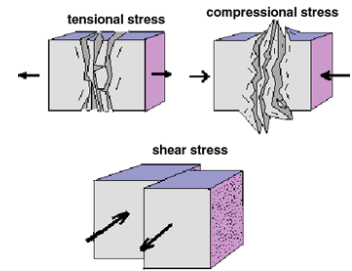
a. Layers of the Earth <http://success.shoreline.edu>

b. **Crust** - outermost layer of the Earth

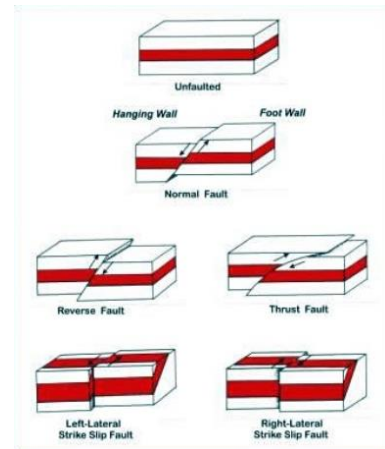
- Continental crust** – make up land masses, 32-70km thick
- Oceanic crust** – make up ocean floor, 8km thick

- c. **Stress** – change shape and volume of the crust, leading to movement and deformation

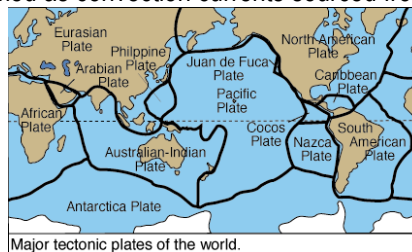
- i. **Compression** - rocks are pushed together causing inc. density and dec. volume
- ii. **Tension** – rocks are pulled apart causing dec. density and inc. volume
- iii. **Shearing** – rocks are pushed in opposite directions causes twisting or break apart



- d. **Fault** – a break or crack along which rocks move (level of fault plane); hanging wall (rock above fault plane) and foot wall (rock below fault plane)
- i. **Normal fault** - hanging wall goes down relative to foot wall (vertical)
 - ii. **Reverse fault** – compression causes hanging wall to move up relative to foot wall
 - iii. **Thrust fault** – compression causes hanging wall to slide over the foot wall
 - iv. **Lateral Strike Slip fault** – one moves to left/right in relation to another (horizontal)

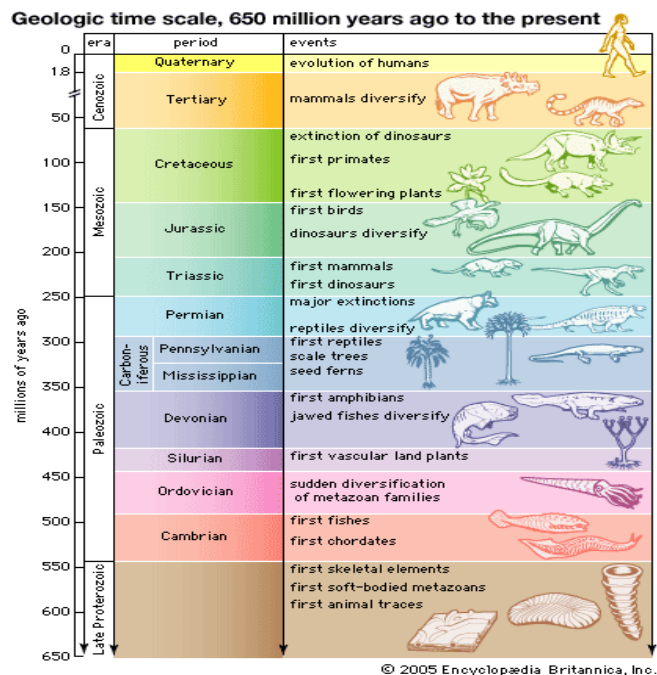


- e. **Plate tectonics**: The Earth's surface is made up of a series of large plates (like pieces of a jigsaw puzzle) which are in constant motion travelling at a few cms per year, spreading from the centre and sinking at the edges, termed as convection currents sourced from radioactive decay of Earth.



- i. **Continental Drift** – theory that continents can drift from one another
Alfred Wegener said that “From supercontinent named Pangea continents moved, the leading edge of the continent would encounter resistance and thus compress and fold upwards forming mountains near the leading edges of the drifting continents”
- ii. **Types of Plate Boundaries**
 1. **Transform boundaries** occur where two lithospheric plates slide or grind past each other along transform faults; plates are neither created nor destroyed. (ex. The San Andreas Fault in California)
 2. **Divergent boundaries** occur where two plates slide apart from each other. (rifting)
 - a. **Ocean-to-ocean rifting**: divergent boundaries form by seafloor spreading, allowing for the formation of new ocean basin. (ex. Mid-Atlantic Ridge, East Pacific Rise)

- b. Continent-to-continent rifting, divergent boundaries may cause new ocean basin to form as the continent spreads, the central rift collapses, and ocean fills the basin. (ex. Africa's East African Rift, Red Sea)
 3. **Convergent boundaries** occur where two plates slide toward each other to form either a subduction zone (one plate moving underneath the other) or a continental collision.
 - a. Continent-to-continent subduction, the dense oceanic lithosphere plunges beneath the less dense continent. Earthquakes then trace the path of the downward-moving plate as it descends into asthenosphere, a trench forms, and as the subducted plate partially melts, magma rises to form continental volcanoes. (ex. Western South America, and Cascade Mountains in Western United States)
 - b. Ocean-to-ocean subduction, older, cooler, denser crust slips beneath less dense crust. This causes earthquakes and a deep trench to form in an arc shape. The upper mantle of the subducted plate then heats and magma rises to form curving chains of volcanic islands. (ex. the Andes mountain range in South America, Aleutian islands, Mariana islands, and the Japanese island arc)
 4. **Plate boundary zones** occur where the effects of the interactions are unclear, and the boundaries, usually occurring along a broad belt, are not well defined and may show various types of movements in different episodes.
3. **Half-life** - is the amount of time required for a quantity to fall to half its value as measured at the beginning of the time period. While the term "half-life" can be used to describe any quantity which follows an exponential decay
4. **Geological Periods**



III. Scientific Method and Experiments

1. **Steps of Scientific Method** - process by which scientists, collectively and over time, endeavor to construct an accurate representation of the world.
 - a. Observation and description of a phenomenon
 - b. Formulation of an hypothesis to explain the phenomena
 - c. Use of the hypothesis to predict the existence of other phenomena, or to predict quantitatively the results of new observations.
 - d. Performance of experimental tests by several independent experimenters and properly designed experiments.
 - i. Confirmation of hypothesis
 - ii. Ruling out of hypothesis
2. Sources of Experimental Error
 - a. **Random error** - error intrinsic to instruments of measurement which has equal probability of producing a measurement higher or lower numerically than the "true" value.
 - b. **Non-random or systematic error** - error due to factors which bias the result in one direction.

***No experiment can be perfectly precise. At the same time, there are standard ways of estimating and reducing errors. It is important to determine the accuracy of a particular measurement and, when stating quantitative results, to quote the measurement error.
3. Hypothesis, Model, Theories, Laws
 - a. **Hypothesis** is a limited statement regarding cause and effect in specific situations; it also refers to our state of knowledge before experimental work has been performed and perhaps even before new phenomena have been predicted.
 - b. **Models** are for situations when it is known that the hypothesis has at least limited validity. (ex. Bohr model)
 - c. **Scientific theory or law** represents a hypothesis, or a group of related hypotheses, which has been confirmed through repeated experimental tests.
4. Variables in an Experiment
 - a. **Independent variable**: factor that you purposely change in an experiment.
 - b. **Dependent variable**: factor that you are measuring in order to determine whether or not it is affected by a change in the independent variable.
 - c. **Controlled variable**: factors which are controlled or held constant throughout an investigation. They are kept unchanging so that they won't influence the outcome of the experiment.

B. Review of Biology

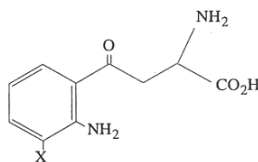
I. Organic Macromolecules

Creation of larger molecules = dehydration (removal of H₂O)

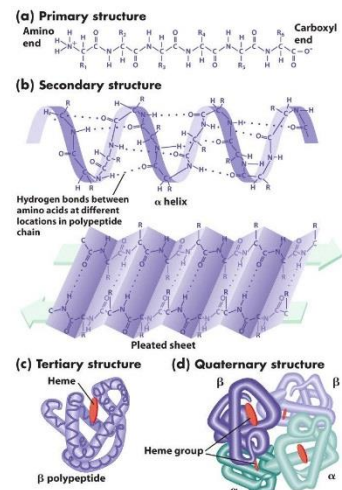
Breakdown of larger molecules = hydrolysis

1. **Proteins** C, H, O, N
 Monomer: Amino Acid (20)
 α helix, β pleated sheets
 Polypeptide chains

Broken down by proteinases

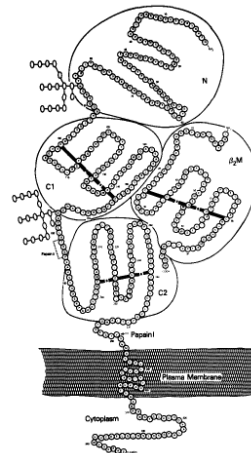


6-7



Found in enzymes, blood cells, receptors and muscle tissue
 created by RNA during DNA Transcription and Translation
 Biuret Test (+) blue → violet (clustering of amide ions)

		Second Position					
		U	C	A	G		
First Position [5' end]	U	UUU } Phe UUC UUA } Leu UUG	UCU } UCC } Ser UCA UCG	UAU } Tyr UAC UAA } Stop UAG } Stop	UGU } Cys UGC UGA } Stop UGG } Trp	U C A G	Third Position [3' end]
	C	CUU } CUC } Leu CUA CUG	CCU } CCC } Pro CCA CCG	CAU } His CAC CAA } Gln CAG	CGU } CGC } Arg CGA CGG	U C A G	
	A	AUU } Ile AUC AUA } Met AUG	ACU } ACC } Thr ACA ACG	AAU } Asn AAC AAA } Lys AAG	AGU } Ser AGC AGA } Arg AGG	U C A G	
	G	GUU } GUC } Val GUA GUG	GCU } GCC } Ala GCA GCG	GAU } Asp GAC GAA } Glu GAG	GGU } GGC } Gly GGA GGG	U C A G	

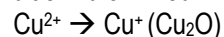


- The protein consists of two polypeptide chains, a long one on the left of 346 amino acids — it is called the heavy chain — and a short one on the right of 99 amino acids.
- The heavy chain has 5 main domains, anchored in the plasma membrane of the cell, the heavy chain is called an integral membrane protein.
 - three **extracellular domains**, designated here as N (includes the N-terminal), C1, and C2
 - transmembrane domain** where the polypeptide chain passes through the plasma membrane of the cell
 - a **cytoplasmic domain** (with the C terminal) within the cytoplasm of the cell.
- To the right is the protein molecule called **beta-2 microglobulin**. It is not attached by noncovalent interactions like hydrogen bonds, and thus called peripheral membrane protein.
- The dark bars represent disulfide (S—S) bridges linking portions of each external domain (except the N domain).

2. Carbohydrates C, H, O [1:2:1]

Monomer: monosaccharide/simple sugar

Benedict's Test (+): blue → brown red

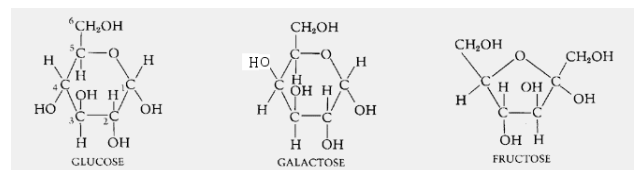


$\text{C}_6\text{H}_{12}\text{O}_6$ = structural isomers

Glucose: blood sugar

Galactose: milk sugar

Fructose: fruit sugar



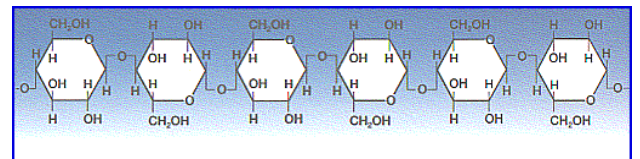
Disaccharide (2 units)

$\text{C}_{12}\text{H}_{22}\text{O}_{11}$ = structural isomers

Sucrose: common table sugar = glucose + fructose

Lactose: major sugar in milk = glucose + galactose

Maltose: product of starch digestion = glucose + glucose



Oligosaccharide (3-10 units)

Polysaccharide/complex sugar

Starch Test (+): black-blue after addition of iodine

- Starch - insoluble in water, plant storage form.
 - Amylose: linear chains of several hundred glucose units linked by a 1-4 glycosidic bond.
 - Amylopectin: highly branched several thousand glucose units. A short side chain is linked to C6 (the carbon above the ring)
 - The hydrolysis of starch is done by amylases.
- Cellulose - single most abundant organic molecule in the biosphere, major structural material of plants. Difference from starch:
 - orientation of the glycosidic bonds: glucose rings arranged in a flip-flop manner → long, straight, rigid molecule.
 - no side chains in cellulose → lie close together
 - many -OH groups → many hydrogen bonds to form
- Glycogen – animal storage form.
 - Structure is similar to amylopectin, although the branches tend to be shorter and more frequent.
 - Stored mostly in liver and muscles
 - Glycogen is broken back down into glucose when energy is needed via glycogenolysis
 - Phosphate groups break the 1 → 4 linkages
 - Then remove phosphate group, so that glucose can leave the cell

3. Lipids C,H,O, [3:1]

3 F.A. chains + glycerol bound by ester linkage = triglyceride

Sudan IV Test (+) = dark red → reddish orange

Hydrophobic

Saturated - all single bonds; maximum possible amount of hydrogens, solid at room temperature

Mono/Polyunsaturated – one/multiple double/triple bond; not bonded to maximum hydrogens, liquid at room temperature

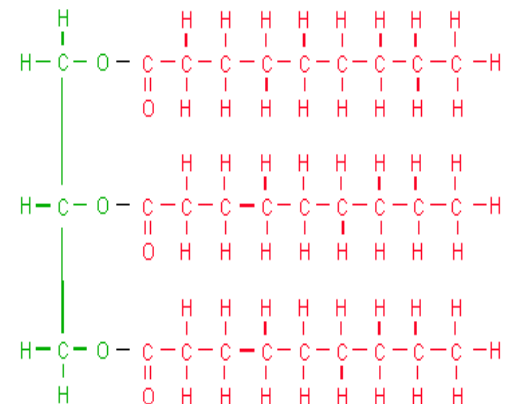
Cis – carbon chains on same side “good oils”

Trans – carbon chains on opposite side “trans fats = bad oils”

Lipoproteins are clusters of proteins and lipids all tangled up together. These act as a means of carrying lipids, including cholesterol, around in our blood.

LDL or low density lipoprotein “bad guy,” associated with deposition of cholesterol

HDL or high density lipoprotein “good guy,” associated with carrying “cholesterol” out of the blood

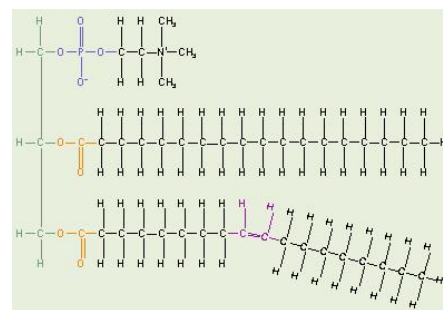


- Triglycerides

Chain Length	# of Carbons	Name	Formula	Notes
The first 11 saturated fatty acids are:				
short-chain	C2	acetic acid	CH_3COOH	<i>aceto</i> = vinegar
	C4	butyric acid	$\text{CH}_3(\text{CH}_2)_2\text{COOH}$	<i>butyr</i> = butter
medium-chain	C6	caproic acid	$\text{CH}_3(\text{CH}_2)_4\text{COOH}$	<i>capri</i> = goat
	C8	caprylic acid	$\text{CH}_3(\text{CH}_2)_6\text{COOH}$	
	C10	capric acid	$\text{CH}_3(\text{CH}_2)_8\text{COOH}$	
	C12	lauric acid	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$	<i>lauri</i> = laurel
long-chain	C14	myristic acid	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$	<i>myrist</i> = anoint, ointment
	C16	palmitic	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	
	C18	stearic acid	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$	<i>stear</i> = fat, suet, tallow
	C20	arachidic or eicosanoic acid	$\text{CH}_3(\text{CH}_2)_{18}\text{COOH}$	<i>arachis</i> = a leguminous plant, present in peanut oil
	C22	behenic or docosanoic acid	$\text{CH}_3(\text{CH}_2)_{20}\text{COOH}$	present peanut & canola oils
	A few of the more important unsaturated fatty acids include:			
EFA = essential fatty acid, PUFA = polyunsaturated fatty acid, MUFA = monounsaturated fatty acid. (CH_2) _n , for example, means repeat CH_2 seven times in a row, so $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2$. To determine the omega number, count the number of carbons in from the left end of each of the following molecules until the first double bond. Do not start counting from the carboxyl group.				
C18	oleic acid	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_9\text{COOH}$	omega-9, MUFA, present in olive & sesame oils, <i>oleo</i> = olive, olive oil	
C18	linoleic acid	$\text{CH}_3(\text{CH}_2)_4\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	EFA, omega-6, PUFA	
C18	linolenic or alpha linolenic acid (ALA)	$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_4\text{COOH}$	EFA, omega-3, PUFA, present in flaxseed oil	
C20	arachidonic acid (ARA)	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_4(\text{CH}_2)_2\text{COOH}$	EFA, omega-6, PUFA	
C20	eicosapentaenoic or timnodonic acid (EPA)	$\text{CH}_3(\text{CH}_2\text{CH}=\text{CH})_5(\text{CH}_2)_3\text{COOH}$	omega-3, PUFA, present in algae/fish oil	
C22	docosahexaenoic or cervonic acid (DHA)	$\text{CH}_3(\text{CH}_2\text{CH}=\text{CH})_6(\text{CH}_2)_2\text{COOH}$	omega-3, PUFA, present in algae/fish oil	
Notice that, while these unsaturated fatty acids contain the same number of carbons as some of the saturated fatty acids, they are chemically different and have different names.				

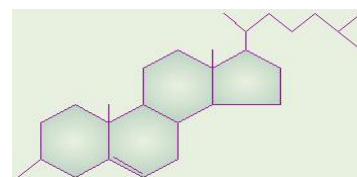
- Phospholipids

- Glycerol + two fatty acids + phosphate group with some other molecule attached to its other end.
- Hydrocarbon tails of the fatty acids are still hydrophobic, but the phosphate group end of the molecule is hydrophilic because of the oxygens with all of their pairs of unshared electrons.
- Soluble in water and oil - "emulsifying agent"
 - Ex. lecithin (found in eggs, used to make mayonnaise), cell membranes



- Steroids

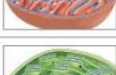
- consist of two six-membered rings side-by-side and sharing one side in common, a third six-membered ring off the top corner of the right ring, and a five-membered ring attached to the right side of that. The central core of this molecule, consisting of four fused rings
- Ex. estrogen (estradiol), progesterone, corticosteroids such as cortisol (cortisone), aldosterone, testosterone, and Vitamin D



II. Cells, Tissues

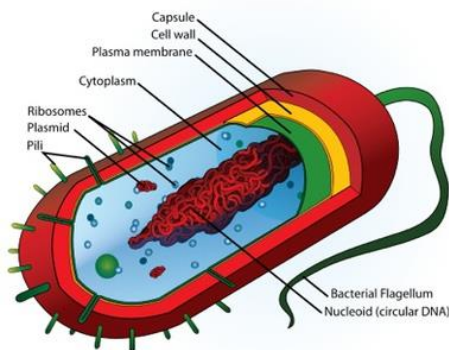
1. Cell components and functions

SUMMARY TABLE 7.2 Eukaryotic Cell Components

Icons not to scale	Structure		Function	
	Membrane	Components		
	Nucleus	Double ("envelope"); openings called nuclear pores	Chromosomes Nucleolus Nuclear lamina	Genetic information Assembly of ribosome subunits Structural support
	Ribosomes	None	Complex of RNA and proteins	Protein synthesis
	Endomembrane system			
	Rough ER	Single; contains receptors for entry of selected proteins	Network of branching sacs Ribosomes associated	Protein synthesis and processing
	Golgi apparatus	Single; contains receptors for products of rough ER	Stack of flattened cisternae	Protein processing (e.g., glycosylation)
	Smooth ER	Single; contains enzymes for synthesizing phospholipids	Network of branching sacs Enzymes for synthesizing lipids	Lipid synthesis
	Lysosomes	Single; contains proton pumps	Acid hydrolases (catalyze hydrolysis reactions)	Digestion and recycling
	Peroxisomes	Single; contains transporters for selected macromolecules	Enzymes that catalyze oxidation reactions Catalase (processes peroxide)	Oxidation of fatty acids, ethanol, or other compounds
	Vacuoles	Single; contains transporters for selected molecules	Varies—pigments, oils, carbohydrates, water, or toxins	Varies—coloration, storage of oils, carbohydrates, water, or toxins
	Mitochondria	Double; inner contains enzymes for ATP production	Enzymes that catalyze oxidation-reduction reactions, ATP synthesis	ATP production
	Chloroplasts	Double; plus membrane-bound sacs in interior	Pigments Enzymes that catalyze oxidation-reduction reactions	Production of ATP and sugars via photosynthesis
	Cytoskeleton	None	Actin filaments Intermediate filaments Microtubules	Structural support; movement of materials; in some species, movement of whole cell
	Plasma membrane	Single; contains transport and receptor proteins	Phospholipid bilayer with transport and receptor proteins	Selective permeability—maintains intracellular environment
	Cell wall	None	Carbohydrate fibers running through carbohydrate or protein matrix	Protection, structural support

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2. Prokaryotic Cell



- **Capsule** - Found in some bacterial cells, this additional outer covering protects the cell when it is engulfed by other organisms, assists in retaining moisture, and helps the cell adhere to surfaces and nutrients.
- **Cell Wall** - Outer covering of most cells that protects the bacterial cell and gives it shape.
- **Cytoplasm** - A gel-like substance composed mainly of water that also contains enzymes, salts, cell components, and various organic molecules.

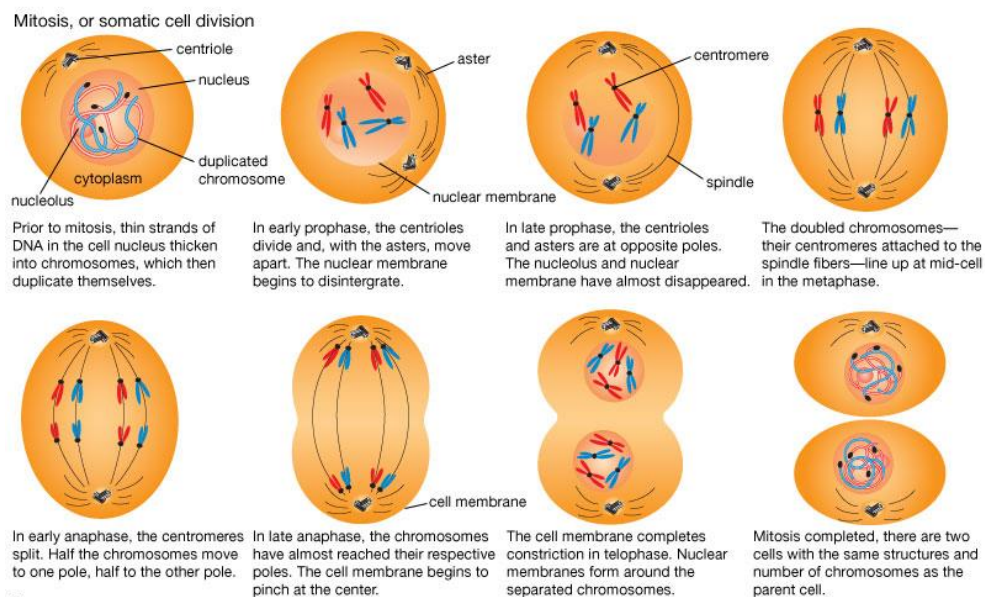
- Cell Membrane or Plasma Membrane - Surrounds the cell's cytoplasm and regulates the flow of substances in and out of the cell.
- Pili - Hair-like structures on the surface of the cell that attach to other bacterial cells. Shorter pili called fimbriae help bacteria attach to surfaces.
- Flagella - Long, whip-like protrusion that aids in cellular locomotion.
- Ribosomes - Cell structures responsible for protein production.
- Plasmids - Gene carrying, circular DNA structures that are not involved in reproduction.
- Nucleoid Region - Area of the cytoplasm that contains the single bacterial DNA molecule.

3. Comparison between Eukaryotic and Prokaryotic Cells

Eukaryotic and Prokaryotic Cell Structures		
Cell Structure	Prokaryotic Cell	Typical Animal Eukaryotic Cell
Cell Membrane	Yes	Yes
Cell Wall	Yes	No
Centrioles	No	Yes
Chromosomes	One long DNA strand	Many
Cilia or Flagella	Yes, simple	Yes, complex
Endoplasmic Reticulum	No	Yes (some exceptions)
Golgi Complex	No	Yes
Lysosomes	No	Common
Mitochondria	No	Yes
Nucleus	No	Yes
Peroxisomes	No	Common
Ribosomes	Yes	Yes

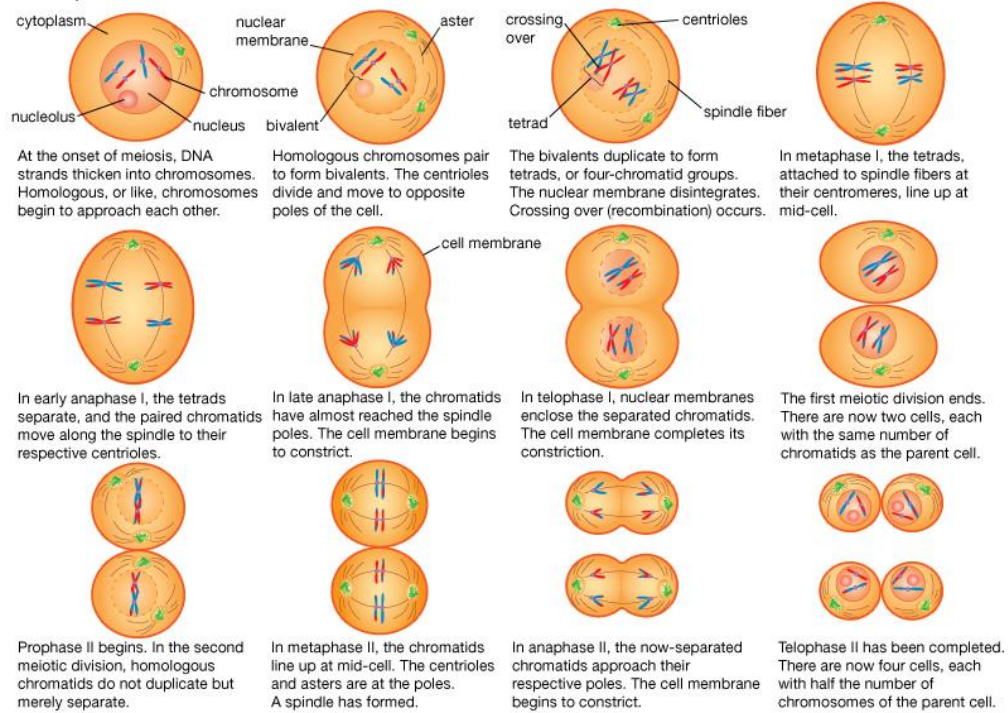
4. Cell Division

a. Mitosis – division of somatic body cells resulting to two daughter cells



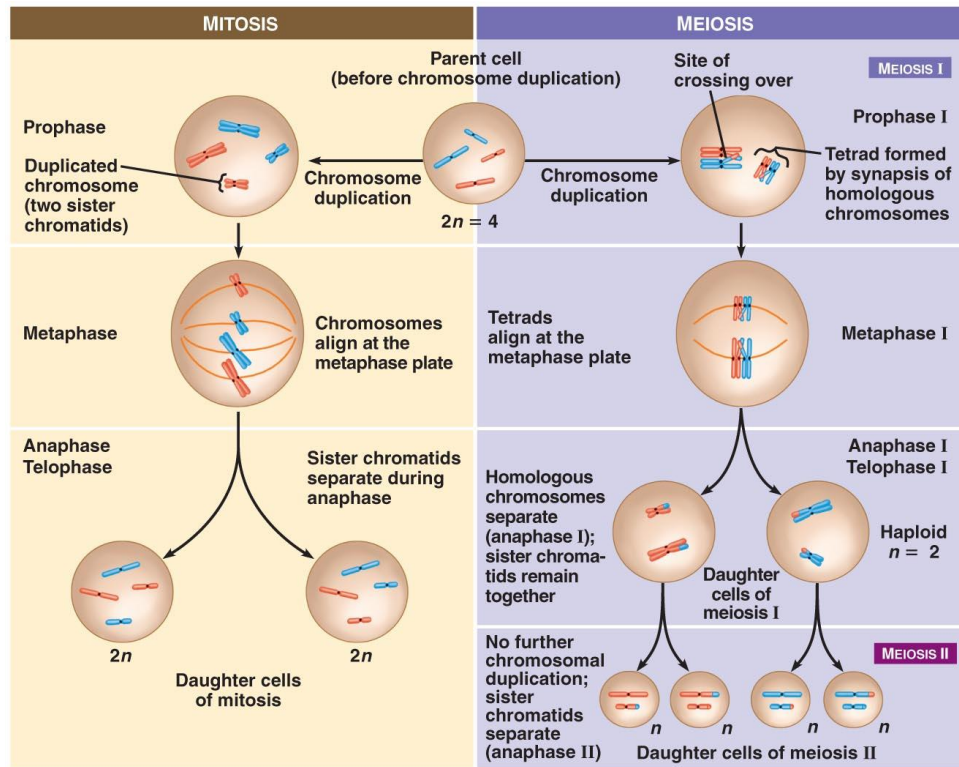
b. Meiosis – division of sex cells/gametes, resulting to 4 recombinant cells

Meiosis, or sex cell division



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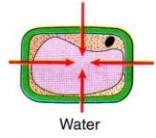
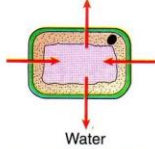
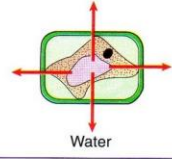
Comparison between Mitosis and Meiosis



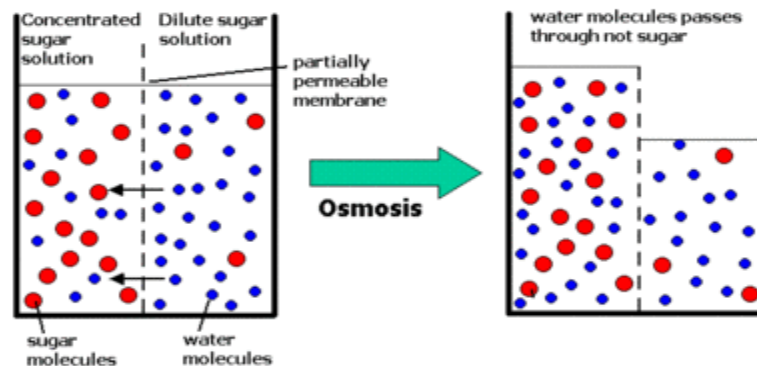
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5. Cell and the Environment

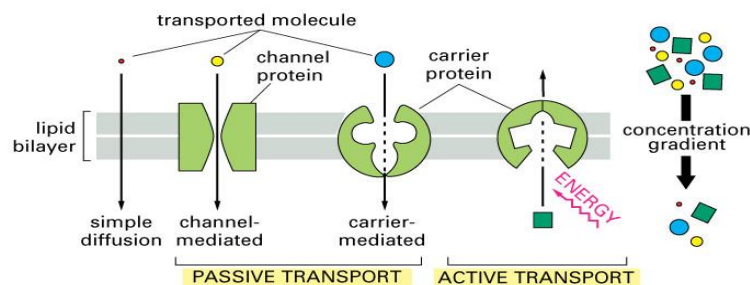
Tonicity – measure of the effective osmotic pressure gradient (measured by the water potential of two solutions) of two solutions separated by a selectively permeable membrane

Types of solution			
	Hypotonic	Isotonic	Hypertonic
Description	• A solution which has less solutes than another solution.	• A solution which has the same amount of solutes with another solution.	• A solution which has more solutes than another solution.
Effect on animal cell	• Water enters the cell. • Cell expands and may finally bursts (Condition known as haemolysis if involves red blood cells). 	• Water molecules move in and out of the cell at same rate. 	• Water moves out of the cell. • Cell shrinks (crenation). 
Effect on plant cell	• Water enters the cell. • Cell becomes very turgid. 	• Water moves in and out of the cell at the same rate. • Cell is turgid (normal). 	• Water moves out of cell. • Cell becomes flaccid (plasmolysis). 

Osmosis - spontaneous net movement of solvent molecules through a partially permeable membrane into a region of higher solute concentration, in the direction that tends to equalize the solute concentrations on the two sides.



Transport of Substances



III. Organ Systems

INTEGUMENTARY SYSTEM

Components: skin, hair, nails, sweat glands

Function: covers and protects body

Specific features:

- Protection against injury
- Sensory receptors in skin tell about the environment (touch, pressure, heat, cold, pain)
- Prevents drying out of cells
- Helps maintain body temperature (capillaries and sweat glands in skin)
- Sweat glands excrete water and some wastes
- Skin has Vitamin D precursor, which is converted to Vitamin D by sunlight

SKELETAL SYSTEM

Components: bones, cartilage, ligaments

Function: supports and protects body; muscles attached to bones; provides calcium storage; site of blood cell formation

Specific features:

- Supports body via bony framework
- Protects delicate vital organs (for their weight, bones are nearly as strong as steel)
- Bones are levers that transmit muscular forces; muscles are attached to bones by bands of connective tissue called tendons. When muscles contract, they pull on bones. Bones are held together at the joints by bands of connective tissue called ligaments.
- Marrow inside some bones produces blood cells (specifically inside flat bones: skull, ribs and breastbone)
- Bones serve as banks for storage and release of minerals like calcium and phosphorus

MUSCULAR SYSTEM

Components: skeletal muscle, cardiac muscle, smooth muscle

Function: moves parts of the skeleton, locomotion; pumps blood; aids movement of internal materials

Specific features:

- Muscle cells contract and become shorter and thicker; because muscle cells are long and narrow, they are called fibers
- Skeletal muscles are attached to bones; they are voluntary muscles, which make our bodies move. They are striped or striated in appearance.
- Cardiac muscle is found in the walls of the heart; it contracts involuntarily and is also striated.
- Smooth muscle is found in the walls of the digestive tract, uterus, blood vessels and other internal organs. The fibers are not striated and they are involuntary.

NERVOUS SYSTEM

Components: nerves and sense organs, brain and spinal cord

Function: receives stimuli from external and internal environments, conducts impulses and integrates activities of other systems

Specific features:

- Two divisions of the nervous system: central and peripheral
- Central nervous system consists of the brain and spinal cord
- Peripheral nervous system consists of the sense organs (eyes, ears, taste buds, olfactory receptors, touch receptors) and the nerves which connect the spinal cord with the rest of the body. These nerves are classified as either afferent (transmit information from the periphery to the spinal cord) or efferent (transmit information from the spinal cord to the periphery).
- Peripheral nervous system is subdivided into two divisions: somatic division (consists

of receptors and nerves concerned with changes in the outside environment; the sense organs and afferent nerves) and the autonomic division (regulates the internal environment; primarily the sympathetic and parasympathetic systems of the efferent system of nerves).

ENDOCRINE SYSTEM

Components: pituitary gland, adrenal gland, thyroid gland, hypothalamus gland, pineal gland, kidney, pancreas, ovaries, testes and other ductless glands (which are defined as tissues that secrete hormones that diffuse into the blood vessels)

Function: regulates body chemistry and many body functions

Specific features:

- These glands are regulated by feedback control: information about hormone levels or their effect is fed back to the gland to regulate the hormone's release
- Endocrine activity is controlled by the hypothalamus gland (which is located in the brain). This gland links the nervous and endocrine systems. As a result of nervous stimuli, it secretes several releasing and inhibiting hormones that affect the activity of the other glands.

CIRCULATORY SYSTEM

Components: heart, blood vessels, blood; lymph and lymph structures

Function: transports materials from one part of the body to another; defends body against disease

Specific features:

- Consists of two subsystems: the cardiovascular system (includes the heart which pumps the blood through the blood vessels) and the lymphatic system (which helps to preserve fluid balance and protects the body against disease)
- Transports nutrients from the digestive system to all parts of the body
- Transports oxygen from the lungs to all the cells of the body
- Transports carbon dioxide and other metabolic wastes from the cells to the excretory organs
- Transports hormones from the endocrine glands to the target tissues
- Helps maintain normal body temperature
- Helps maintain fluid balance
- Protects the body against disease-causing organisms. The lymphocytes, which are a type of white blood cell, are formed in the lymph tissue: lymph glands, spleen, thymus, tonsils and lymphoid tissue in the gut. There are two types of lymphocytes: T lymphocytes (the mediators of cellular immunity; these cells destroy the invader) and B lymphocytes (the antibody-producing cells; humoral immunity).

RESPIRATORY SYSTEM

Components: lungs and air passageways

Function: exchanges gases between the blood and the external environment

Specific features:

- Respiration includes breathing, gas exchange between lungs and blood, transport of gases through the body by the blood, gas exchange between the blood and the cells and cellular respiration (the chemical reaction pathways by which chemical energy is obtained from food).

DIGESTIVE SYSTEM

Components: mouth, esophagus, stomach, intestines, liver, pancreas

Function: ingests and digests foods, absorbs them into the blood

Specific features:

- Salivary glands, liver and pancreas are not part of the digestive system but secrete digestive juices into it
- The digestive system involves four major processes:
 1. Ingestion-taking food into the mouth, chewing and swallowing
 2. Digestion-breakdown of food into smaller pieces (catalyzed by enzymes)
 3. Absorption-transfer of digested food through the wall of the intestine and into the circulatory system

4. Elimination-removal of undigested and unabsorbed food from the body (in feces)

URINARY SYSTEM

Components: kidney, bladder and associated ducts

Function: excretes metabolic wastes; removes substances present in excess from the blood

Specific features:

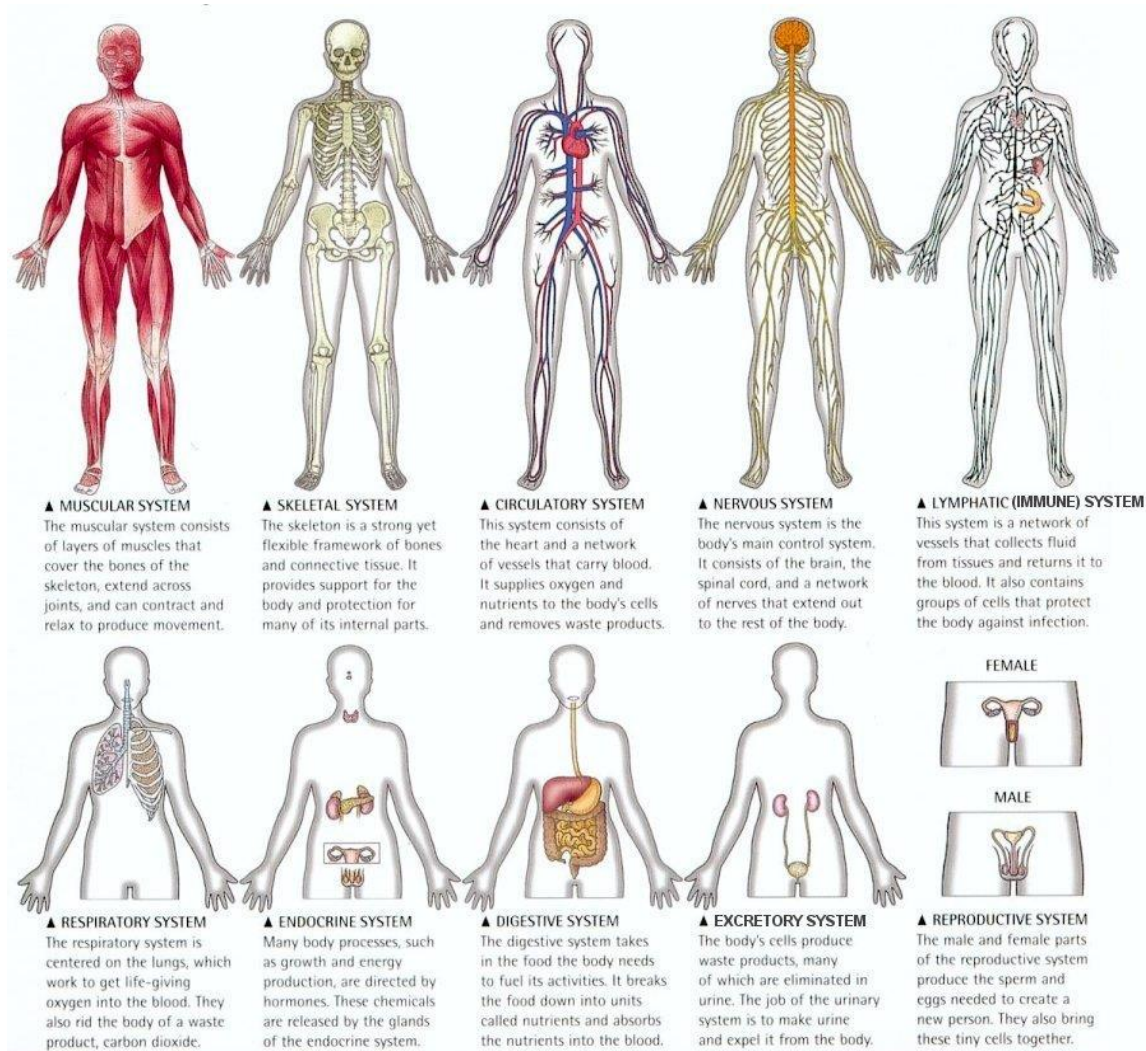
-Urine is made by the kidneys; it's transported from the kidneys to the bladder by the ureters; the bladder stores the urine then the urine leaves the bladder and exits the body via the urethra.

-95% of urine is water. Also present is urea, which is produced in the liver (urea is the excretion form of nitrogen waste).

REPRODUCTIVE SYSTEM

Components: testes, ovaries and associated structures

Function: reproduction, which provides for continuation of the species



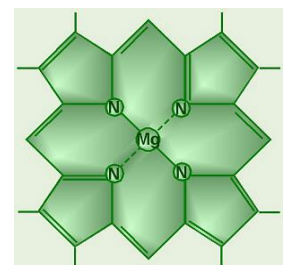
IV. Energy

1. **Photosynthesis:** Photosynthesis is the process of converting light energy to chemical energy and storing it in the bonds of sugar. This process occurs in plants and some algae (Kingdom Protista). Plants need only light energy, CO₂, and H₂O to make sugar. The process of photosynthesis takes place in the chloroplasts, specifically using chlorophyll, the green pigment involved in photosynthesis.

1. Chloroplast: outer and inner membranes, intermembrane space, stroma, and thylakoids stacked in grana. The chlorophyll is built into the membranes of the thylakoids.
2. $6\text{CO}_2 + 6\text{H}_2\text{O} (+ \text{light energy}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

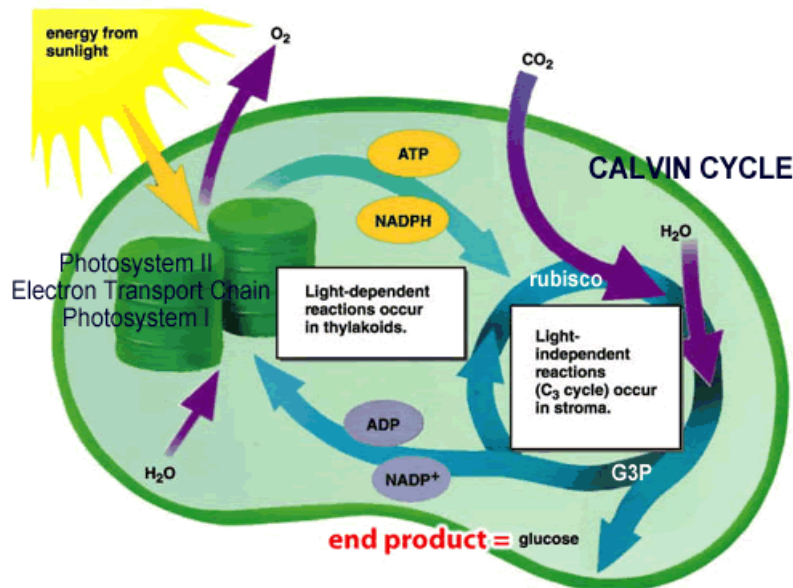
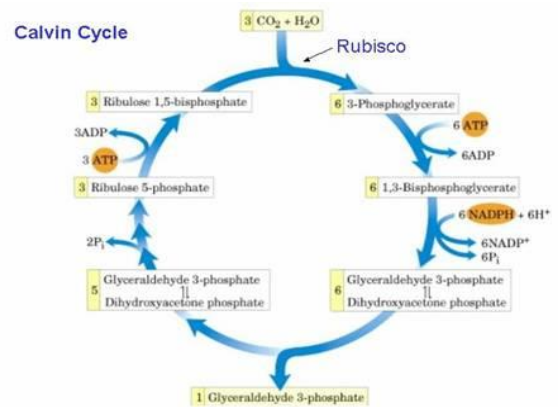
a. Light Reaction

- happens in the thylakoid membrane, converts light energy to chemical energy.
- chlorophyll and several other pigments such as beta-carotene are organized in clusters in the thylakoid membrane. Each of these differently-colored pigments can absorb a different color of light and pass its energy to the central chlorophyll molecule to do photosynthesis.
- central part of the chemical structure of a chlorophyll molecule is a porphyrin ring, which consists of several fused rings of carbon and nitrogen with a magnesium ion in the center.



b. Dark Reaction

- the energy harvested via the light reaction is stored by forming ATP (adenosine triphosphate)
- takes place in the stroma within the chloroplast, and converts CO₂ to sugar.
- needs the products of the light reaction (ATP and NADPH).
- the first product of photosynthesis is a three-carbon compound glyceraldehyde 3-phosphate. Almost immediately, two of these join to form a glucose molecule.
- Rubisco - Ribulose-1,5-bisphosphate carboxylase/oxygenase
 - atmospheric carbon dioxide is converted by plants to energy-rich glucose. By catalyzes the carboxylation of ribulose-1,5-bisphosphate (RuBP).
 - most abundant protein on Earth



2. Cell Respiration: a catabolic pathway for the production of adenosine triphosphate (ATP)

- Glycolysis:** "splitting sugars." Glucose, a 6C sugar, is split into two molecules of a 3C sugar. In the process, two molecules of ATP, two molecules of pyruvic acid and two NADH are produced. Glycolysis can occur with or without oxygen.
 - (+) oxygen, glycolysis is the first stage of cellular respiration.
 - (-) oxygen, glycolysis allows cells to make small amounts of ATP via fermentation.

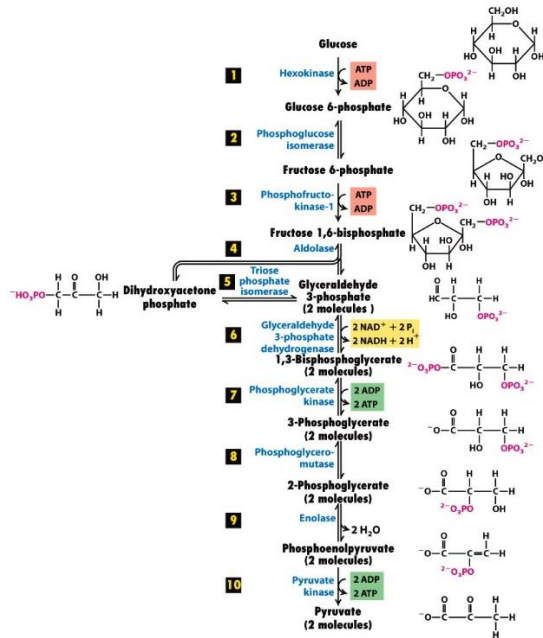
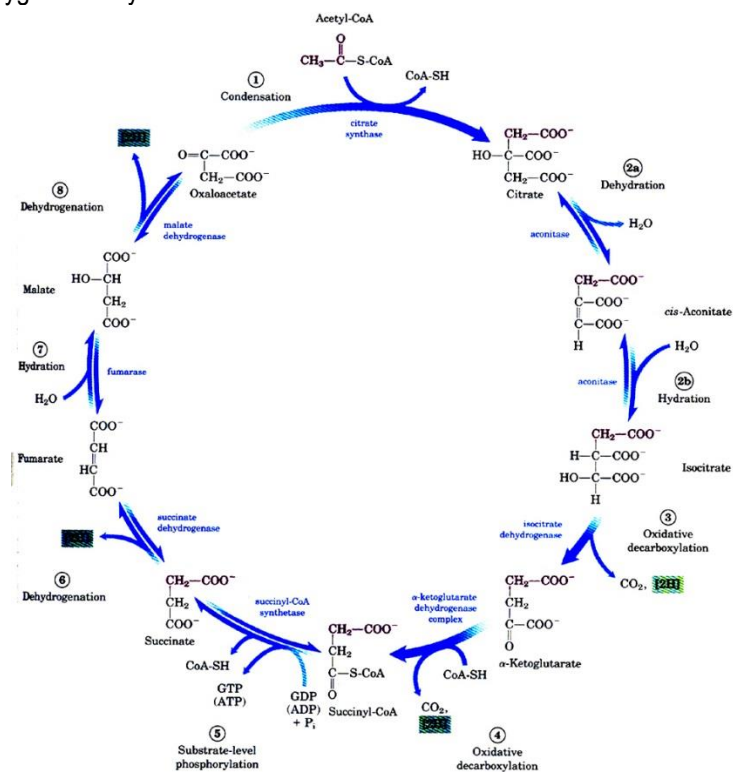
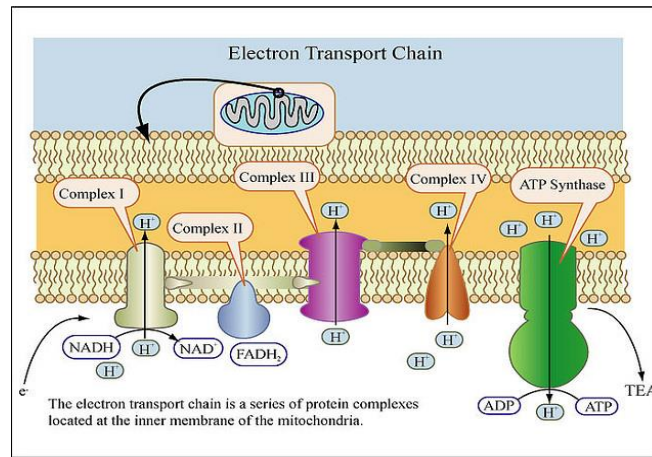


Figure 12-3
Molecular Cell Biology, Sixth Edition
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2. **Citric Acid Cycle:** aka Krebs Cycle begins after the two molecules of 3C sugar produced in glycolysis are converted to acetyl CoA. Through a series of intermediate steps, several compounds capable of storing "high energy" electrons are produced along with two ATP molecules. These compounds, known as nicotinamide adenine dinucleotide (NAD) and flavin adenine dinucleotide (FAD), are reduced in the process. The Citric Acid Cycle occurs only when oxygen is present but it doesn't use oxygen directly.



3. **Electron Transport:** requires oxygen directly. The electron transport chain is a series of electron carriers in the membrane of the mitochondria in eukaryotic cells. Through a series of reactions, the "high energy" electrons are passed to oxygen. In the process, a electric gradient is formed, and ATP is produced.

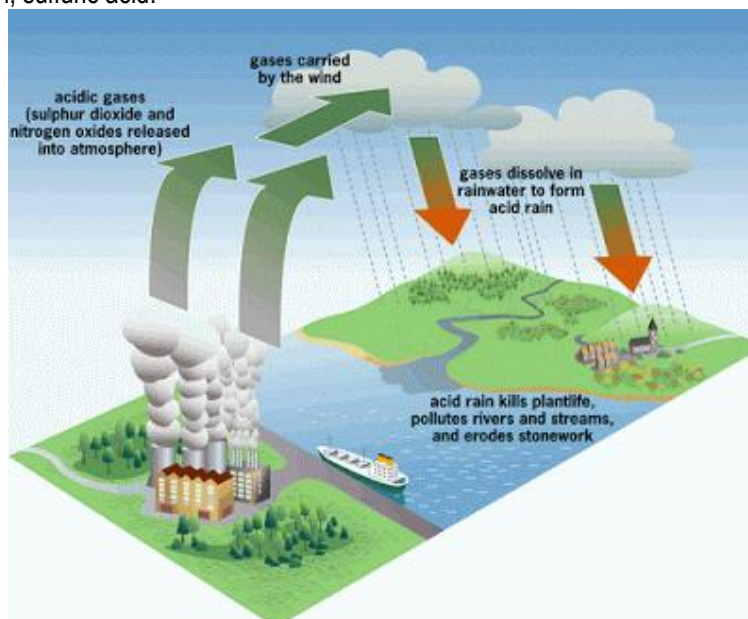


4. **Maximum ATP Yields:** In summary, prokaryotic cells can yield a maximum of **38 ATP molecules** while eukaryotic cells can yield a maximum of **36 ATP**. In eukaryotic cells, the NADH molecules produced in glycolysis pass through the mitochondrial membrane, which needs two ATPs.

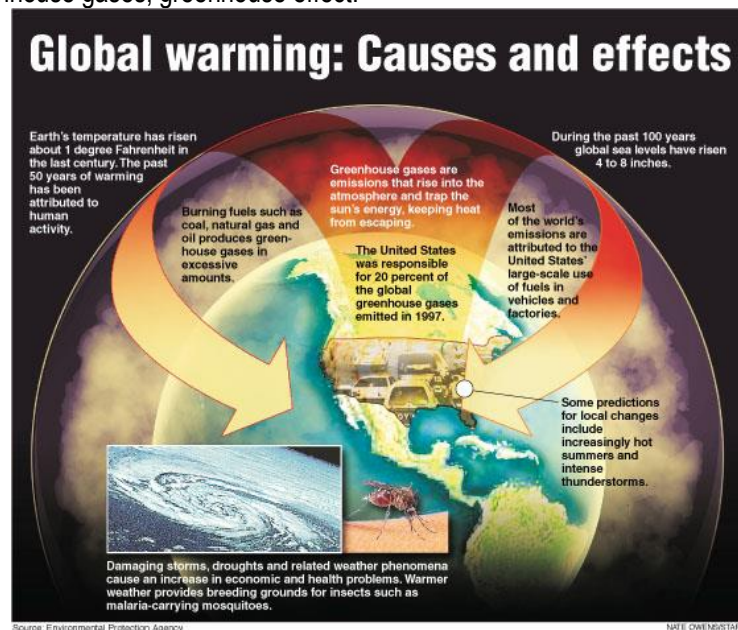
V. Ecology and Environment\

ENVIRONMENTAL SCIENCE

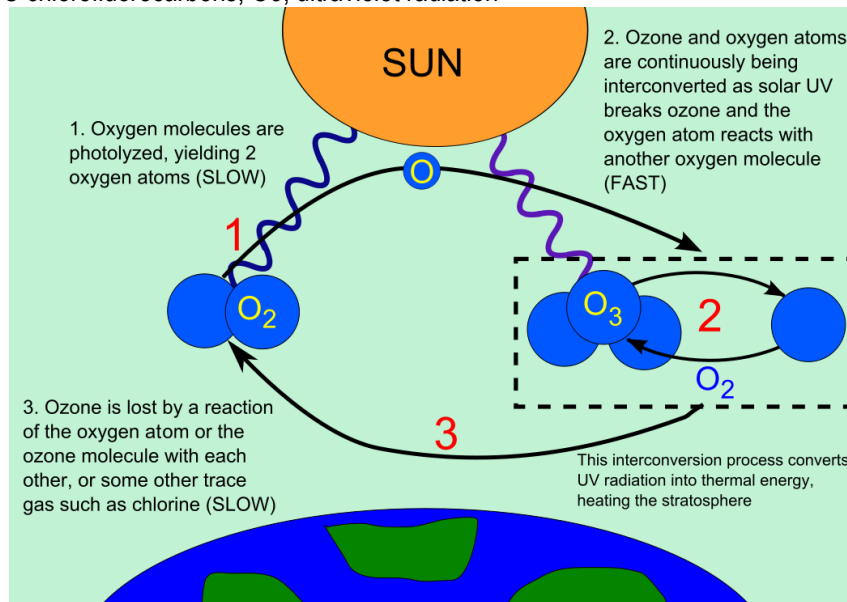
Acid Rain - air pollution, sulfuric acid.



Global Warming - greenhouse gases, greenhouse effect.



Ozone Layer - CFC chlorofluorocarbons, O₃, ultraviolet radiation



Loss of Biodiversity - deforestation, extinction of species.

Toxic Magnification - accumulation of toxins in higher trophic levels.

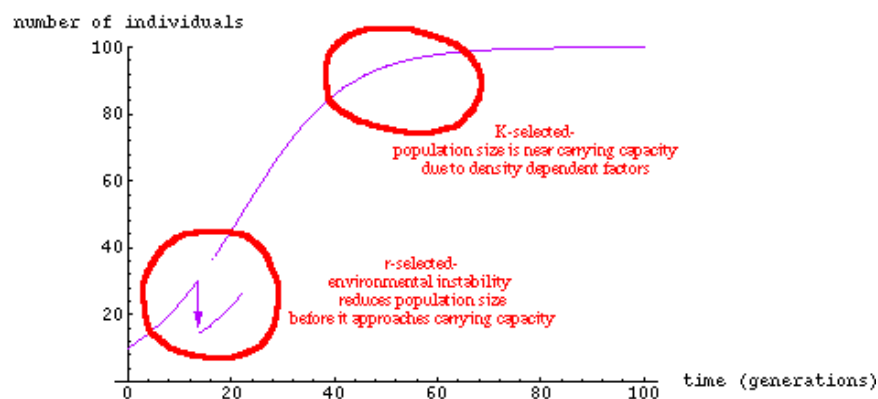
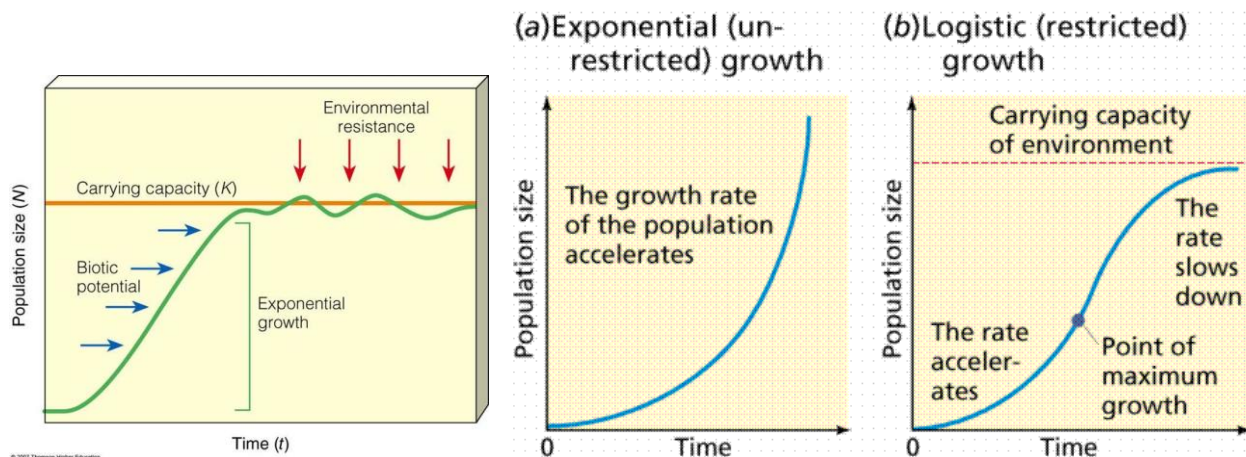
Resource Depletion - topsoil erosion, ground water pollution and depletion, non-renewable resources, rate of renewable resources.

Human Population Growth - growth rate, ecological footprint.

POPULATION ECOLOGY

Populations - size, density, dispersion.

Population Growth Model - exponential growth, logistic growth, carrying capacity, density-dependent factors, density-independent factors, r-strategists, K-strategists.

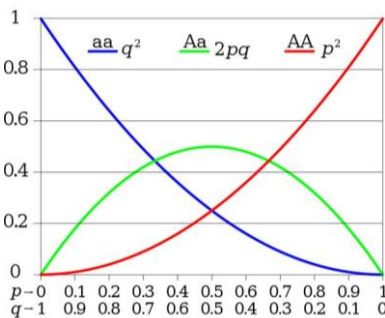


Hardy-Weinberg principle - allele frequencies in populations

Assumptions of Hardy-Weinberg

1. No mutations
2. Population large
3. Population is isolated
4. All members survive & reproduce
5. Random mating

These essentially never occur in nature, so Hardy-Weinberg is an *ideal case*



The Hardy-Weinberg Principle

$$p^2 + 2pq + q^2 = 1$$

frequency of homozygous dominant genotype frequency of heterozygous genotype frequency of homozygous recessive genotype

BIOLOGICAL COMMUNITY - the different species living in a habitat.

Ecosystem Biodiversity - biotic factors and abiotic factors

Competition - niche, limited resources.

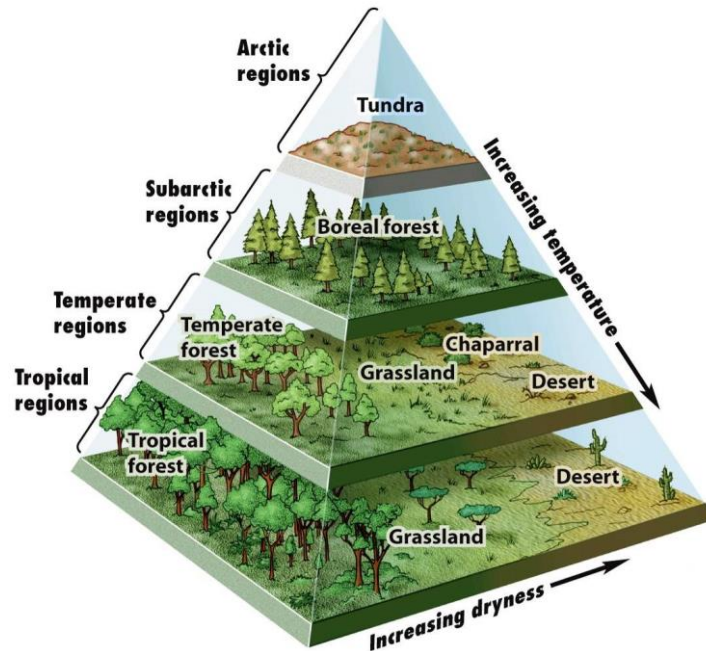
Symbiosis - parasitism, mutualism, commensalism, predator and prey

Biomes of the World (defined as contiguous areas with similar climatic conditions on the Earth, such as communities of plants, animals, and soil organisms, and are often referred to as ecosystems)

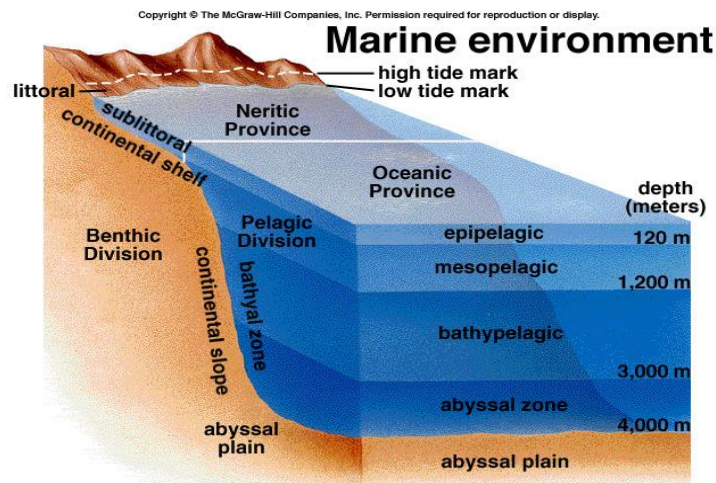
tundra biome, desert biome, marine biome, grassland biome, taiga biome, temperate grassland biomes, temperate deciduous forest biome, tropical rain forest biome, savanna biome, polar ice biome.

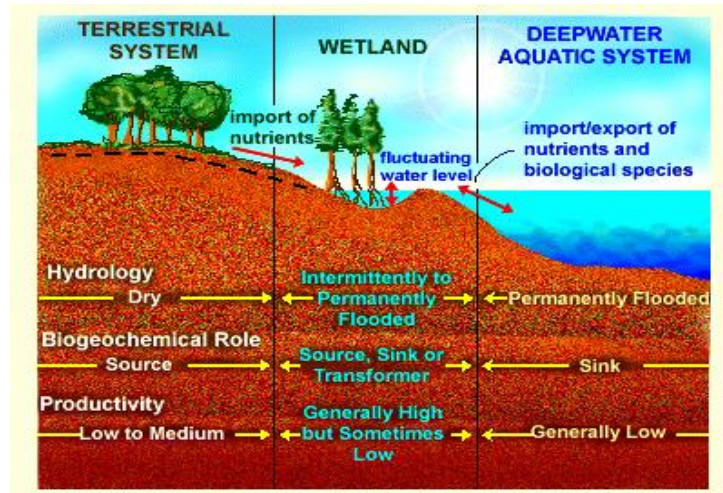
What is an Ecosystem - biological community and physical environmental factors.

Forest Ecosystems - tropical rainforest ecosystems, temperate rainforest, tropical dry forests, deciduous forests, coniferous forests.



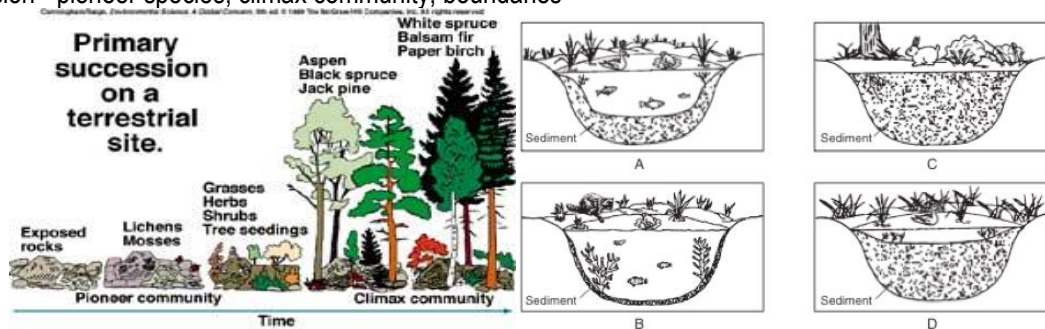
Aquatic Ecosystems - freshwater, ocean, wetland, marine, pond





Other Ecosystems - desert, tundra, grassland, soil

Succession - pioneer species, climax community, boundaries



ENERGY FLOW

Food Chain - feed from lower trophic level.

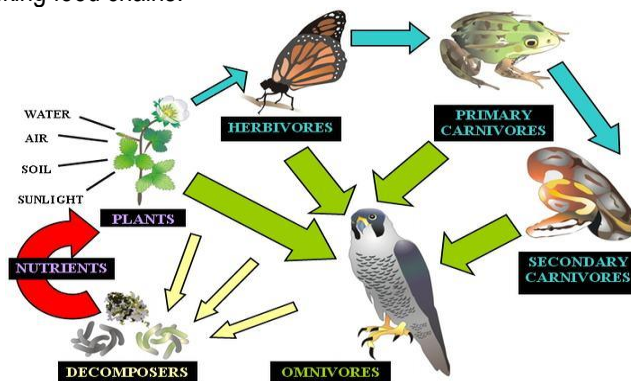
Trophic Levels - producers, consumers, herbivores, carnivores, omnivores, detritivores, decomposers.

Autotrophs (producers) vs. Heterotrophs (consumers)

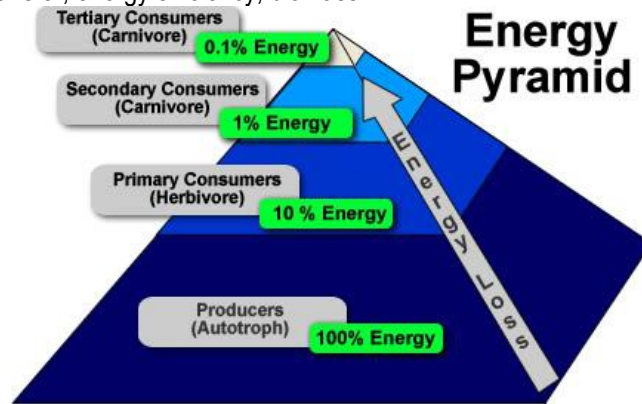
Poikilotherms (ectotherms) vs. Homeotherms (endotherms)



Food Web - multiple interlocking food chains.

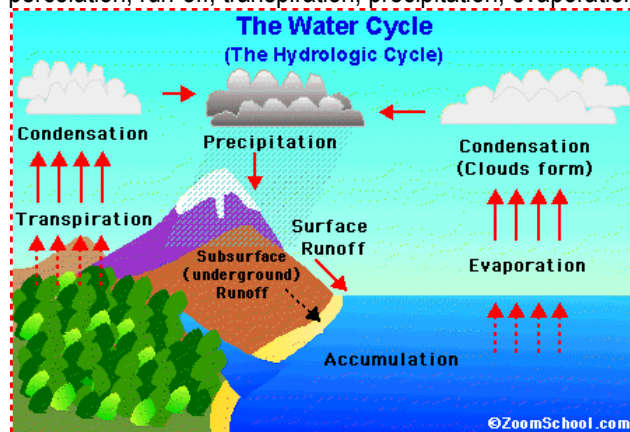


Energy Pyramid - energy transfer, energy efficiency, biomass.

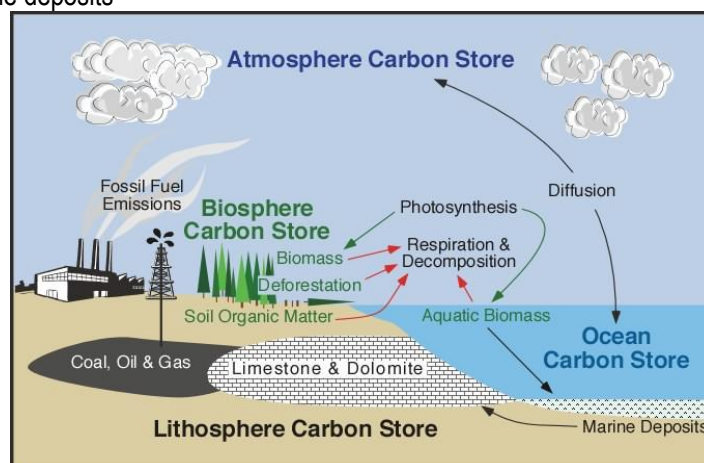


BIOGEOCHEMICAL CYCLES

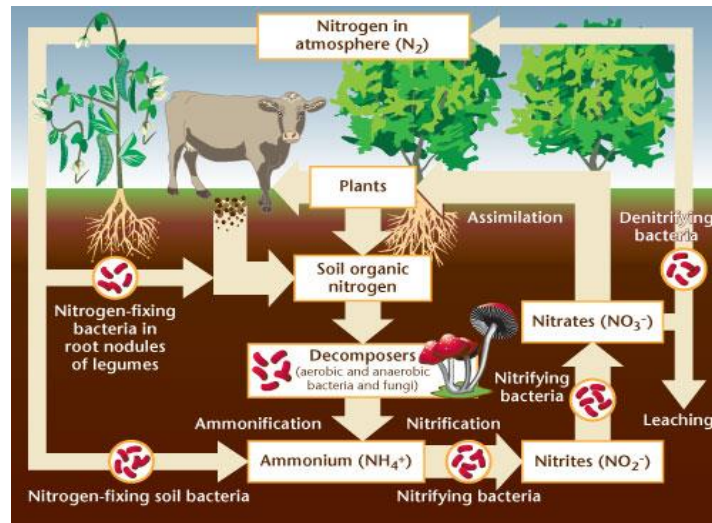
Water Cycle - ground water, percolation, run-off, transpiration, precipitation, evaporation, water vapor



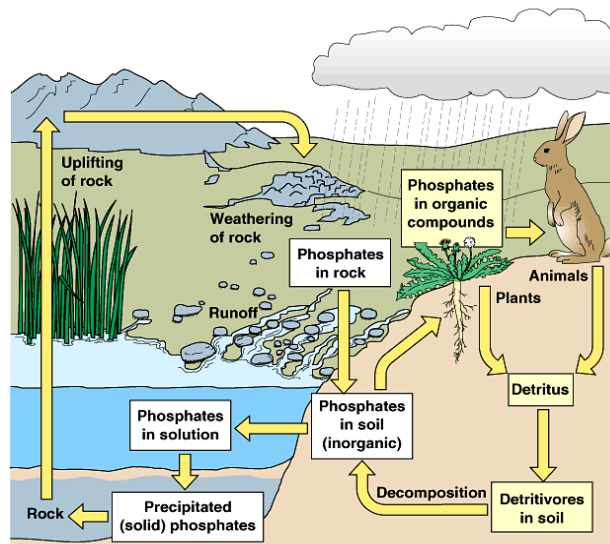
Carbon Cycle - atmospheric CO₂, cellular respiration, photosynthesis, combustion, death and decomposition, fossil fuels, erosion from marine deposits



Nitrogen Cycle - nitrogen fixation, nitrogen fixing bacteria, assimilation, ammonification, nitrification, nitrifying bacteria, denitrification, denitrifying bacteria, atmospheric N₂, ammonia, nitrates, nitrites.

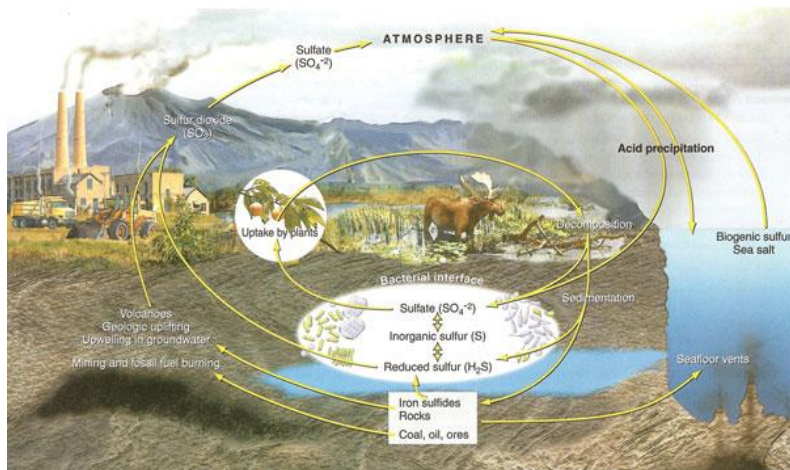


Phosphorous Cycle



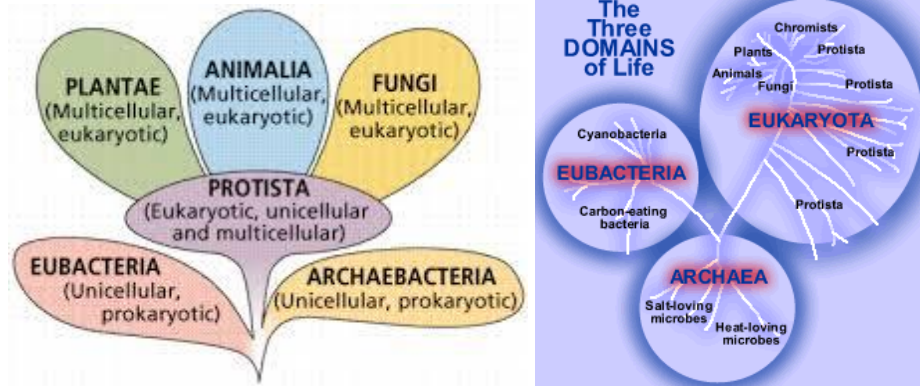
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Sulfur Cycle



VI. Taxonomy – binomial nomenclature

Kingdom, Phylum, Class, Order, Family, Genus, and Species.



1. Archaeobacteria

Organisms: Methanogens, Halophiles, Thermophiles, Psychrophiles

Cell Type: Prokaryotic

Metabolism: Depending on species - oxygen, hydrogen, carbon dioxide, sulfur, sulfide may be needed for metabolism.

Nutrition Acquisition: Depending on species - nutrition intake may be by absorption, non-photosynthetic photophosphorylation, or chemosynthesis.

Reproduction: Asexual reproduction by binary fission, budding, or fragmentation.

2. Eubacteria

Organisms: Bacteria, Cyanobacteria(blue-green algae), Actinobacteria

Cell Type: Prokaryotic

Metabolism: Depending on species - oxygen may be toxic, tolerated, or needed for metabolism.

Nutrition Acquisition: Depending on species - nutrition intake may be by absorption, photosynthesis, or chemosynthesis.

Reproduction: Asexual reproduction

3. Protista

Organisms: Amoebae, green algae, brown algae, diatoms, euglena, slime molds

Cell Type: Eukaryotic

Metabolism: Oxygen is needed for metabolism.

Nutrition Acquisition: Depending on species - nutrition intake may be by absorption, photosynthesis, or ingestion.

Reproduction: Mostly asexual reproduction. Meiosis occurs in some species.

4. Fungi

Organisms: Mushrooms, yeast, molds

Cell Type: Eukaryotic

Metabolism: Oxygen is needed for metabolism.

Nutrition Acquisition: Absorption

Reproduction: Asexual or sexual reproduction occur.

5. Plantae

Organisms: Mosses, angiosperms (flowering plants), gymnosperms, liverworts, ferns

Cell Type: Eukaryotic

Metabolism: Oxygen is needed for metabolism.

Nutrition Acquisition: Photosynthesis

Reproduction: Some species reproduce asexually by mitosis. Other species exhibit sexual reproduction.

6. Animalia

Organisms: Mammals, amphibians, sponges, insects, worms

Cell Type: Eukaryotic

Metabolism: Oxygen is needed for metabolism.

Nutrition Acquisition: Ingestion

Reproduction: Sexual reproduction

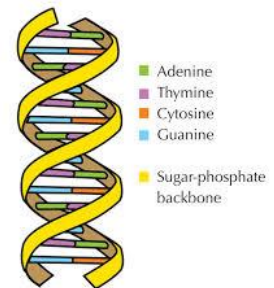
VII. Molecular Biology and Genetics

1. **Nucleic Acids:** allow organisms to transfer genetic information from one generation to the next. Two types of nucleic acids:

- a. ribonucleic acid (RNA) CGAU
- b. deoxyribonucleic acid (DNA) CGAT

Nucleic acids: monomer – nucleotides composed of

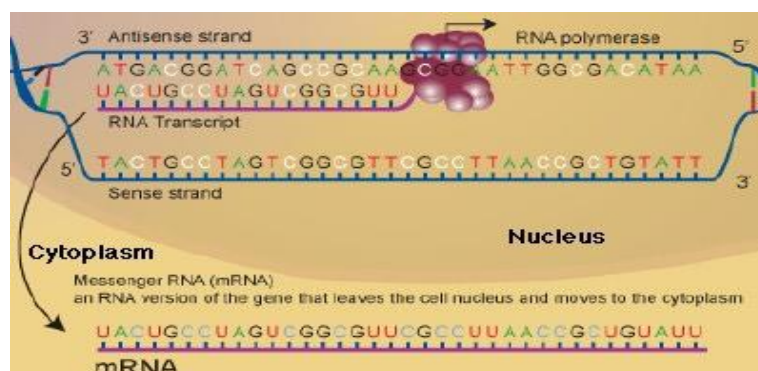
- Nitrogenous Base – GA (purine) CTU (pyrimidine) complementary pairs
- Five-Carbon Sugar – ribose, deoxyribose
- Phosphate Group



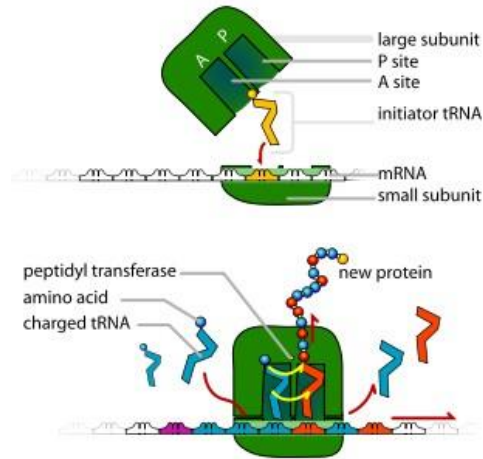
Polynucleotides: nucleotides are joined to one another by covalent bonds between the phosphate of one and the sugar of another, called phosphodiester linkages.

2. **Central Dogma of Molecular Biology:** When a cell divides, its DNA is copied and passed from one cell generation to the next generation. DNA contains the "programmatic instructions" for cellular activities. When organisms produce offspring, these instructions, in the form of DNA, are passed down. RNA is involved in the synthesis of proteins. "Information" is typically passed from DNA to RNA to the resulting proteins.

- a. DNA Transcription - transcribing DNA to RNA. The RNA transcript is used to produce proteins.
 - **RNA Polymerase** Binds to DNA: DNA is transcribed by RNA polymerase. Specific nucleotide sequences tell RNA polymerase where to begin and where to end. RNA polymerase attaches to the DNA at the promoter region.
 - **Elongation:** Certain proteins called transcription factors unwind the DNA strand and allow RNA polymerase to transcribe only a single strand of DNA into a single stranded RNA polymer called *messenger RNA (mRNA)*. The strand that serves as the template is called the antisense strand. The strand that is not transcribed is called the sense strand.
 - **Termination:** RNA polymerase moves along the DNA until it reaches a terminator sequence. At that point, RNA polymerase releases the mRNA polymer and detaches from the DNA. mRNA must cross the nuclear membrane to reach the cytoplasm.



- b. Translation - Once in the cytoplasm, ribosomes and another RNA molecule called transfer RNA work together to translate mRNA into a protein. Proteins can be manufactured in large quantities because a single DNA sequence can be transcribed by many RNA polymerase molecules at once.



3. Mendelian Genetics

MENDEL'S CONCLUSIONS BASED ON HIS EXPERIMENTS:

- The hereditary determinants are of a particulate nature. These determinants are called genes.
- Each parent has a gene pair in each cell for each trait studied. The F1 from a cross of two pure lines contains one allele for the dominant phenotype and one for the recessive phenotype. These two alleles comprise the gene pair.
- One member of the gene pair segregates into a gamete, thus each gamete only carries one member of the gene pair.
- Gametes unite at random and irrespective of the other gene pairs involved.

DEFINITIONS

- **Allele** - one alternative form of a given allelic pair; tall and dwarf are the alleles for the height of a pea plant; more than two alleles can exist for any specific gene, but only two of them will be found within any individual
- **Allelic pair** - the combination of two alleles which comprise the gene pair
- **Homozygote** - an individual which contains only one allele at the allelic pair; for example DD is homozygous dominant and dd is homozygous recessive; pure lines are homozygous for the gene of interest
- **Heterozygote** - an individual which contains one of each member of the gene pair; for example the Dd heterozygote
- **Genotype** - the specific allelic combination for a certain gene or set of genes
- **Dominance** - the ability of one allele to express its phenotype at the expense of an alternate allele; the major form of interaction between alleles; generally the dominant allele will make a gene product that the recessive can not; therefore the dominant allele will express itself whenever it is present
- **Pure Line** - a population that breeds true for a particular trait [this was an important innovation because any non-pure (segregating) generation would and did confuse the results of genetic experiments]
- **Phenotype** - literally means "the form that is shown"; it is the outward, physical appearance of a particular trait

- **Dominant** - the allele that expresses itself at the expense of an alternate allele; the phenotype that is expressed in the F1 generation from the cross of two pure lines
- **Recessive** - an allele whose expression is suppressed in the presence of a dominant allele; the phenotype that disappears in the F1 generation from the cross of two pure lines and reappears in the F2 generation

MENDELIAN LAWS:

Mendel's First Law - the law of segregation; during gamete formation each member of the allelic pair separates from the other member to form the genetic constitution of the gamete.

Mendel's Second Law - the law of independent assortment; during gamete formation the segregation of the alleles of one allelic pair is independent of the segregation of the alleles of another allelic pair.

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BIOLOGY

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